Clive Rafiq

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Education

Arizona State University — Tempe, AZ
B.S.E. in Computer Systems Engineering — *Expected May 2028*

- **Coursework:** OOP & Data Structures, Computer Organization & Assembly, C/C++ Programming, Discrete Math

Experience

Backend Developer Intern — May 2026 – Aug 2026
World Salon — *New York, NY*

- Traced 2 production-blocking failures with CloudWatch and container-level debugging (an ASGI adapter failing every request under Python 3.12, and JWKS auth fetches breaking across container boundaries), unblocking production both times
- Shipped per-recipient read/unread tracking for a serverless notification service (FastAPI, AWS Lambda, DynamoDB), designing the per-user state model, PATCH endpoint and query filter so each recipient can mark notifications read and filter to unread ones
- Wrote a reusable Bash harness that publishes SQS events and polls DynamoDB to terminal state, exposing a race condition that reported in-flight notifications as successful and verifying ingestion, conditional-write idempotency and malformed-event isolation

Software Engineer Intern, Backend — Dec 2025 – Jan 2026
BFI Finance Indonesia — *Jakarta, Indonesia*

- Built an event-driven Go microservice (NATS, internal APIs) that caches WhatsApp customer photos and videos in object storage behind URL references, extending retention from a 2-week limit to indefinite
- Deployed it with Docker and Kubernetes into the production loan workflow used daily by 6,000 loan officers serving about 400K customers

Software Engineer Intern, Full-Stack — Jul 2024 – Jan 2025
PhysioActive Indonesia — *Jakarta, Indonesia*

- Automated sentiment analysis with Python ETL pipelines in BigQuery that classified 10,000+ customer messages with an LLM, giving the CEO same-day results that changed how the company talks to customers
- Launched a Python (Flask) web dashboard surfacing customer messages and performance metrics; test messages sent before and after launch showed employee response time down 77%

Leadership

Co-Founder | *TechTrek Indonesia* — Aug 2023 – Feb 2025

- Founded a STEM education nonprofit in Jakarta, running Python and VEX Robotics workshops for 30 local students

Projects

SlackDB | *FastAPI, Slack Bolt, Docker, PostgreSQL* | **AutoDB Track Winner**, VillageHacks ASU — Apr 2026

- Co-built a Slack bot bringing text-to-SQL, migration risk analysis, and query optimization into team channels
- Designed guardrails for LLM-generated SQL: a multi-stage approval flow with risk-scored migrations, sandbox validation, rollback detection, and approver enforcement to block destructive production changes
- Shipped the FastAPI backend (async HTTPX, Slack Bolt, interactive Block Kit) via Docker on Railway with a full audit log

Drift | *React, MediaPipe, WebSerial, ESP32/C++* | *StarkHacks (Purdue)* — Apr 2026

- Built a passive focus tracker fusing typing telemetry (WPM, pause and backspace rate), MediaPipe gaze and blink detection, and ambient noise/light into weighted Focus and Fatigue scores, calibrated against a 30-second per-user baseline
- Wrote a bidirectional WebSerial protocol and embedded C++ firmware for the ESP32, learned during the hackathon, streaming sensor readings to the browser and rendering the computed focus score back to an onboard OLED

Rician Denoising with Deep Learning for MRI Tumor Segmentation | *Python, FFDNet (PyTorch)* — Aug – Dec 2025

- Adapted FFDNet to remove Rician noise from roughly 3,000 MRI slices, improving PSNR by ∼11 dB and cutting mean squared error from ∼851 to ∼81
- Built the training and evaluation pipeline: preprocessing, noise-level conditioning, and per-class benchmarking

Technical Skills

Programming Languages: Python, Go, C, C++, Java, JavaScript, SQL, Bash
Frameworks & Infra: FastAPI, Flask, React, NATS, Docker, Kubernetes, Terraform, AWS (Lambda, SQS, DynamoDB, API Gateway, CloudWatch), Linux, Git, CI/CD (GitHub Actions), AI-Assisted Development
Data & ML: PostgreSQL, Redis, MongoDB, BigQuery, PyTorch, Pandas, NumPy, NLP & LLM Pipelines
Concepts: Distributed Systems, Unit & Integration Testing, Embedded Systems, Data Structures & Algorithms